OWASP Juice Shop — Lab Environment Setup

1. Objective

To install and configure OWASP Juice Shop locally to establish a controlled environment for practicing web application security testing, vulnerability identification, and penetration testing techniques.

2. Prerequisites

- Node.js and npm installed on the system.
- Git installed for cloning the repository.
- A web browser to access the application.

3. Installation and Configuration

Step 1: Install Node.js
Installed a compatible version of Node.js and npm by following the official OWASP Juice Shop Node.js compatibility guide.

Step 2: Clone the Repository
Executed the following command to download the OWASP Juice Shop source code:
git clone https://github.com/juice-shop/juice-shop.git --depth 1
The --depth 1 option downloads only the latest commit history, reducing the amount of data retrieved.

Step 3: Navigate to the Project Directory
Change the path to: **cd juice-shop**

Step 4: Install Dependencies

Executed the following command to install the required project dependencies: **npm install**
This command installs the packages required to run the application. It is generally needed for the initial setup and when project dependencies change.

Step 5: Start the Application
This command starts the OWASP Juice Shop application locally: **npm start**

Step 6: Access the Application
Opened the following URL in a web browser: **http://localhost:3000**
Verified that the application loaded successfully and was ready for security testing.

4. Security Testing Scope

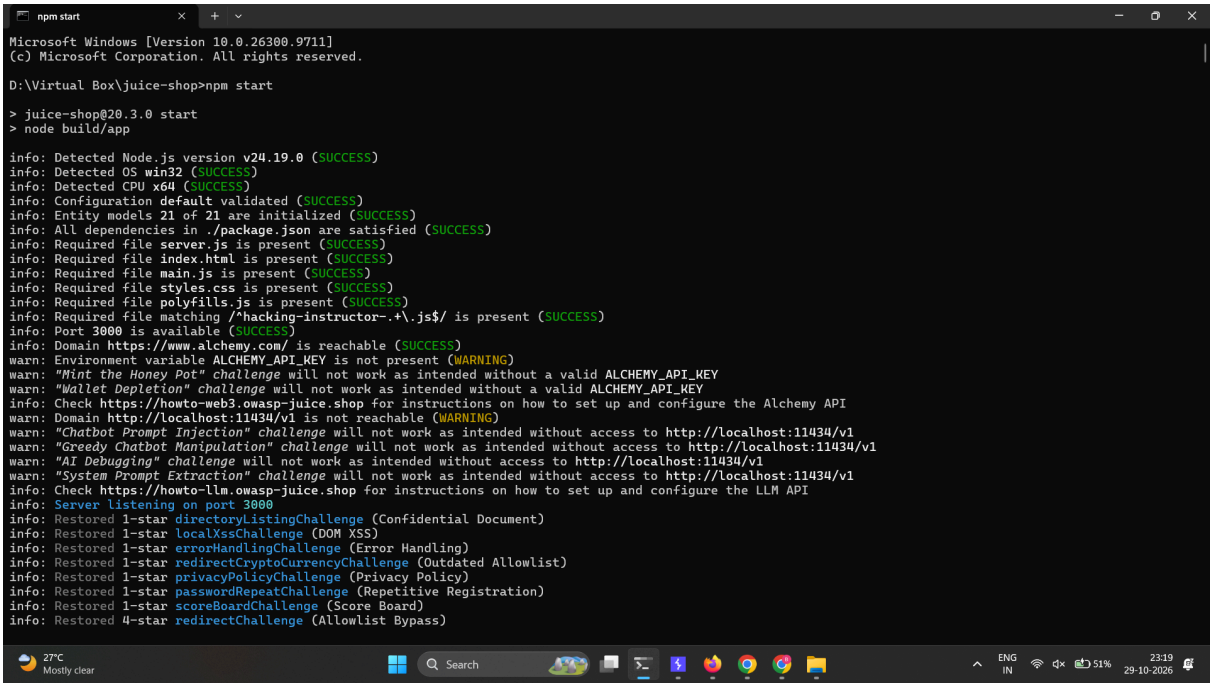
After setting up the application, used it as a controlled lab environment to practice web application reconnaissance, directory and route discovery, vulnerability analysis, and security testing using tools such as DIRB, dirsearch, and OWASP ZAP.
All testing was performed against the locally hosted training application.

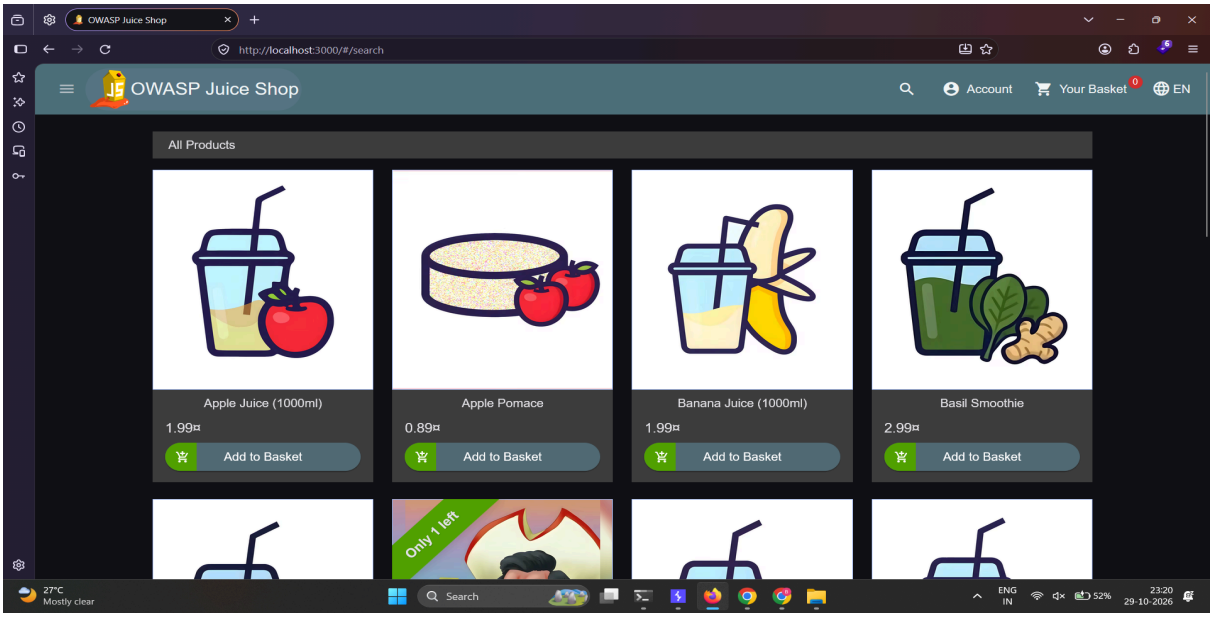
5. Result

Successfully installed and configured OWASP Juice Shop using its GitHub repository and Node.js environment. The application provided a local environment for performing authorized security testing and documenting findings, potential impacts, and remediation recommendations. **Official Repository:** https://github.com/juice-shop/juice-shop

6. Testing Environment

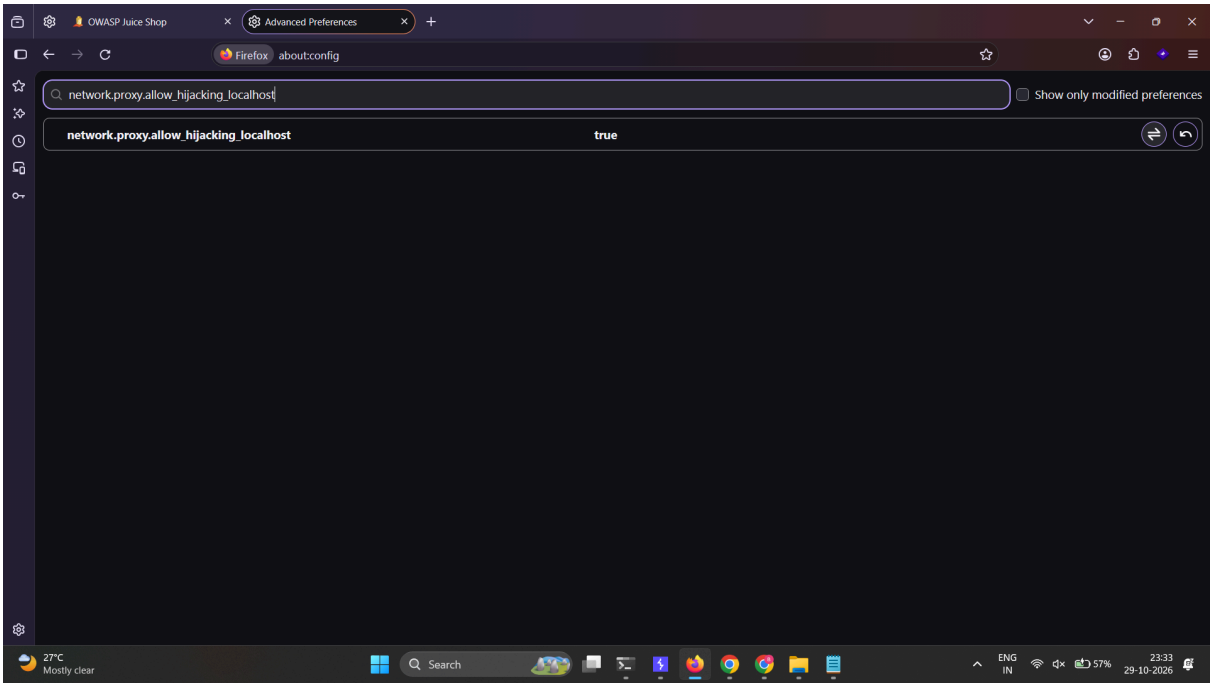
Component	Configuration
Application	OWASP Juice Shop
Installation Method	Git clone and npm
Runtime	Node.js
Package Manager	npm
Access URL	http://localhost:3000
Testing Environment	Localhost / isolated lab





Mozilla Firefox advanced configuration setting used when testing web applications on localhost with security tools such as Burp Suite or OWASP ZAP.

It is especially useful when you're learning web application penetration testing.



How to enable it in Firefox

1. Open Mozilla Firefox.
2. Type **about:config** in the address bar and press Enter.
3. Accept the warning if prompted.
4. Search for **network.proxy.allow_hijacking_localhost.**
5. Change its value from **false to true.**
6. Configure Firefox's proxy settings to match your proxy listener. For example, Burp Suite commonly uses 127.0.0.1:8080.

It enables when you need to intercept local traffic through your configured proxy, so you can inspect traffic requests and responses.